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Graph neural network-based drug-drug interaction prediction

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With the growing variety of pharmacological compounds and the increasing need for polypharmacy, accurately predicting drug-drug interactions (DDIs) is essential to ensure both treatment efficacy and patient safety. Beneficial DDIs can enhance therapeutic outcomes. In contrast, adverse interactions may result in toxicity, reduced efficacy, or even fatality. Thus, the accurate prediction of DDIs is paramount. Building on recent advancements in graph neural network (GNN) architectures, this paper extends prior research, such as the SAGE GNN model, Graph Attention Network model, and Graph Diffusion Network model, by integrating advanced techniques such as skip connections, post-processing layers, and optimized training methods. It start from basic GNN to buld more advanced models such as based on Adaptive Graph Diffusion model. Our experimental results shows based on evaluation on 3 different drug-drug interaction dataset that on some evaluation metric basic models outperforms the advanced ones. We have found that GCN with skip connections, GCN with NGNN and SAGE with NGNN give competent accuracy with other baseline models. The corresponding code and datasets used in this study are available on GitHub for reproducibility at: https://github.com/khushnood/DrugDruginteractionPredictionBasedOnGNN.

When multiple drugs are taken simultaneously, the activity and effectiveness of one drug can be affected by another substance, usually another drug. Drug interactions are categorized into synergism and antagonism. Synergism refers to the combined use of different drugs that enhances their therapeutic effect, while antagonism occurs when two drugs interact, leading to side effects such as reduced efficacy and adverse reactions in patients. In fact, adverse reactions caused by drug interactions are quite common. For example, in recent years, many antiallergic drugs, such as loratadine and diphenhydramine, have been combined with macrolides. Serious adverse reactions can occur after the combined use of antibiotics (e.g., erythromycin). According to relevant studies, gastrointestinal reactions and skin allergies are the most common adverse effects of various macrolide antibiotics, with incidence rates as high as 48% and 26%, respectively[1]. Therefore, researchers are actively focusing on developing computational approaches to identify such interactions. The figure 1 depicts how the drug interaction prediction problem is converted to graph ML-based prediction problem.

Research background

In recent years, the use of polypharmacy–administering multiple drugs simultaneously–has become a common approach for managing complex diseases such as cancer. This strategy is particularly advantageous for older adults with multiple comorbidities, as it utilizes the synergistic effects of drug combinations to enhance treatment efficacy. However, unintended drug-drug interactions (DDIs) remain a significant concern, posing risks such as adverse side effects and drug toxicity. With the growing demand for multidrug therapies, the need for reliable DDI detection methods has become increasingly urgent. Traditional in vitro and in vivo approaches for identifying DDIs are both expensive and time-consuming, particularly given the vast number of possible drug combinations. Consequently, developing effective strategies for DDI prediction has become a priority in pharmaceutical research. Proactively identifying potential DDIs not only minimizes the occurrence of adverse drug reactions (ADRs) but also streamlines the drug development process, ultimately improving patient safety and therapeutic outcomes.

Currently, the variety of drug interactions is vast, yet only a limited number of DDIs are identified through clinical trials and chemical experiments. As a result, some drug interactions remain unnoticed until they are widely used on the market. Often, only after patients experience adverse reactions and report them are these interactions discovered. Although this reactive approach does uncover harmful drug interactions, it comes at

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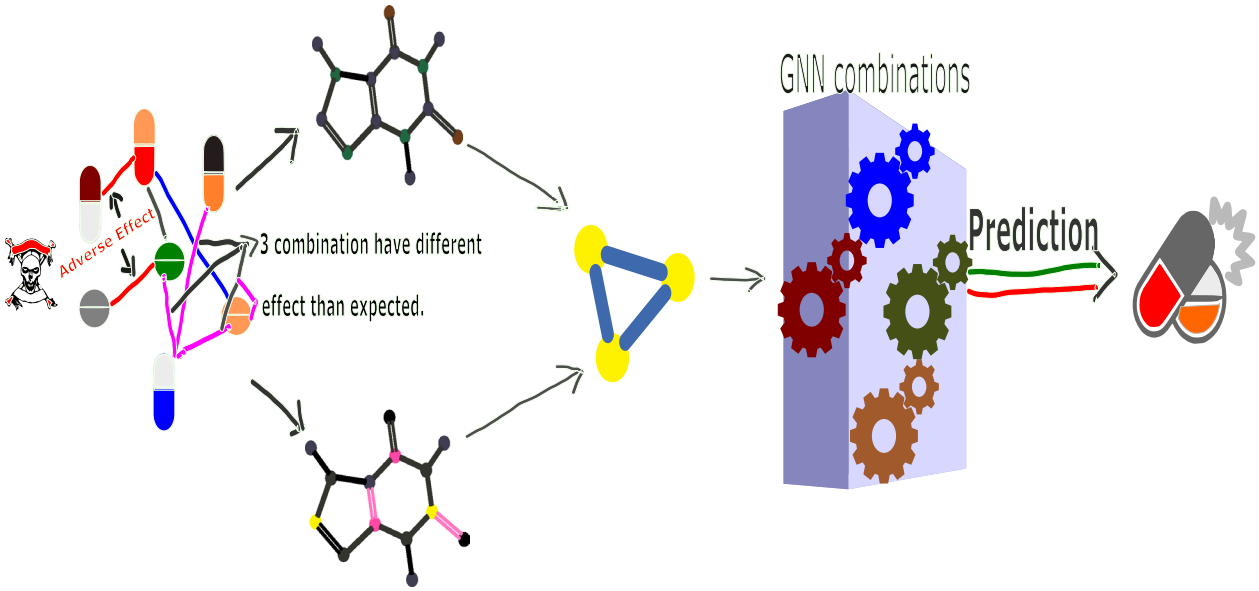


Fig. 1. The figure illustrates the process of converting a drug molecule into graph representations, followed by the application of a Graph Neural Network (GNN) to make predictions.

the cost of patient health and safety, making it an unsustainable solution. Therefore, there is an urgent need for more proactive approaches to predict drug interactions.

In recent years, the accumulation of biological and medical data has led to the widespread application of machine learning in the prediction of DDI. Machine learning algorithms are now being used to automatically extract features from datasets, converting them into biologically meaningful vectors or graph-based data. These features are then analyzed using machine learning models to predict potential drug interactions. Currently, advances in the analysis of graph data structures have led to the development of sophisticated techniques to study graph-based data. This has drawn significant attention to the use of machine learning in this field. One notable method is the calculation of the weighted average of neighboring node information using neural networks, a technique that forms the basis of graph neural networks (GNNs). GNNs have become a powerful tool for processing graph data structures, making them highly suitable for DDI prediction.

Research status

In June 2021, Zhu Jie, Wang Jingxiang, and colleagues introduced a drug efficacy prediction system (DLEPS) based on genetic fingerprinting and deep learning[2]. This study employed neural networks to establish connections between compound structures and gene expression profiles, successfully predicting drug interactions and validating these predictions using real clinical datasets. In early 2022, Wang Fei, Lei Xiujuan, Liao Bo, and Wu Fangxiang introduced a multi-kernel graph convolutional network (GCNMK) for drug interaction prediction[3]. The GCNMK model incorporates two distinct DDI kernels within the graph convolutional layer: one based on an "increasing" DDI graph that captures positive DDI correlations, and the other based on a "decreasing" DDI graph that captures negative DDI correlations. The model demonstrated higher prediction accuracy, with the learned drug features being fed into a block of three fully connected layers for DDI prediction. Gao et al.[4] introduced AutoDDI, an automated method for predicting drug-drug interactions (DDIs). This approach leverages a reinforcement learning-based search algorithm to design optimal GNN architectures without the need for manual intervention. AutoDDI explores a customized search space of handcrafted GNN models, ensuring efficient and accurate predictions. The method achieves state-of-the-art performance on real-world datasets, demonstrating its effectiveness. Additionally, AutoDDI improves interpretability by accurately identifying the drug substructures involved in DDIs. Geng et al.[5] came up with a graph neural network-based model that leverages a multiscale graph neural network (MGNN) and an attention-based substructure interaction learning module to predict adverse drug-drug interactions accurately. MGDDI addresses existing limitations by effectively capturing molecular substructure variations and detailed interactions between drug pairs, demonstrating superior predictive performance in experimental evaluations. Feng et al.[6] proposed a graph attention network-based model for drug-drug interaction (DDI) prediction, designed to handle both known drugs within the DDI network and novel drugs. By constructing a five-layer graph attention network that captures k-hop low-dimensional feature representations and employing an MLP predictor, GNN-DDI achieves superior predictive performance. Additionally, it enhances interpretability by identifying specific drug substructures involved in DDIs, providing insights into their underlying mechanisms. Similarly, Han et al.[7] introduced SmileGNN, a novel DDI prediction model that integrates structural features from SMILES data with topological features derived from knowledge graphs using graph neural networks. Experimental results show that SmileGNN outperforms existing models in DDI prediction accuracy, and its credibility is further validated by verifying five of the top ten predicted interactions with the latest database.

Liu et al.[8] developed DANN-DDI, a deep attention neural network-based framework for predicting missing drug-drug interactions (DDIs). This model integrates multiple drug feature networks and utilizes graph embedding to learn meaningful drug representations, while an attention mechanism is employed to enhance the learning of interactions between drug pairs. Experimental results indicate that DANN-DDI is capable of identifying novel DDIs and associated events, thus improving the robustness and applicability of DDI prediction. In a similar vein, Li et al.[9] introduced HetDDI, a pre-trained heterogeneous graph neural network model designed to enhance DDI prediction. HetDDI combines structural information from drug molecule graphs with rich semantic data from biomedical knowledge graphs, enabling it to effectively learn drug representations from diverse sources of heterogeneous information. Case studies further demonstrate HetDDI's capacity to predict previously unknown DDIs, confirming its practical applicability in real-world scenarios. Sukanya et al.[10] proposed an innovative method for DDI prediction that leverages rich drug representations sourced from multiple knowledge domains, such as the Drug-Target Interaction Network and chemical structure data. This method, which incorporates techniques like metapath2vec and Graph Auto-Encoders, has been shown to outperform previous models across several benchmark datasets. Shtar et al.[11] introduced two novel methods, Adjacency Matrix Factorization (AMF) and Adjacency Matrix Factorization with Propagation (AMFP), for predicting DDIs. These methods frame the DDI prediction problem as link prediction and utilize artificial neural networks and factor propagation to uncover interactions. Lei et al.[12] presented the Neighborhood Relation-Aware Graph Neural Network (NRAGNN), which improves DDI prediction by incorporating both the topological structure of drug entities and the semantic significance of relationships within the drug knowledge graph. The NRAGNN model showed superior performance on the KEGG-DRUG dataset, achieving significant improvements in various evaluation metrics compared to existing models. Lastly, Feng et al.[13] introduced DPDDI, a method for predicting DDIs that uses a Graph Convolution Network (GCN) to extract network structure features from a DDI network, while a Deep Neural Network (DNN) is employed to perform the prediction. DPDDI outperformed four existing methods, demonstrating the effectiveness of GCN-derived features and the concatenation feature aggregation operator in capturing complex DDI information.

Khaled et al.[14] introduce a novel model called the Hypergraph Neural Network (HyGNN) that predicts Drug-Drug Interactions (DDIs) by leveraging only SMILES strings. This model captures the chemical structure similarities of drugs through hypergraph representations, highlighting the importance of structural relationships in drug interactions. Similarly, Junpeng et al.[15] present the Multi-layer Adaptive Soft Mask Graph Neural Network (MASMDDI) model, which enhances DDI prediction by capturing interaction information between chemical substructures. The effectiveness of MASMDDI was evaluated using the DrugBank dataset, where it demonstrated promising results in predicting DDIs for previously unknown drugs. In a different approach, Wang et al.[16] introduce AttenSyn, an attention-based deep graph neural network designed for predicting synergistic drug combinations. This model leverages molecular graphs along with an attention-based pooling module to refine the representations of drug pairs, significantly outperforming current methods for anticancer drug synergy prediction. Additionally, AttenSyn provides interpretability by identifying the key drug substructures contributing to synergy, and further validation on two cell lines demonstrates its ability to generalize across different biological contexts. Wu et al.[17] present MKG-FENN, a multimodal knowledge graph fused end-to-end neural network aimed at DDI prediction. MKG-FENN constructs a multimodal knowledge graph and extracts high-order semantic features through a graph neural network, which are then fused in an end-to-end manner, enabling more robust DDI predictions. Yu et al.[18] propose the Substructure-aware Tensor Neural Network (STNN-DDI), which improves the prediction of multiple types of DDIs by mapping drugs into a substructure-substructure interaction (SSI) space. This model not only enhances predictive performance but also offers interpretability by identifying the critical substructure pairs that drive interactions. Deepa et al.[19] introduce the SNF-HCNN framework, which leverages drug data from sources like DrugBank, PubChem, and SIDER. Using a similarity matrix derived from drug features, SNF-HCNN offers a novel approach to predicting DDIs. Similarly, Su et al.[20] introduce DDKG, an attention-based knowledge graph representation learning framework that integrates drug attributes and neighboring information to enhance DDI predictions. Feng et al.[21] present deepMDDI, a deep learning-based model for multi-label DDI prediction. By utilizing a relational graph convolutional network encoder and a tensor-like decoder, deepMDDI performs well in both transductive and inductive settings, successfully identifying new valid DDIs in case studies. Lastly, Wang et al.[22] propose MIRACLE, a multi-view graph contrastive representation learning method that captures both inter-view molecular structures and intra-view drug interaction relationships, significantly improving DDI prediction accuracy. Further DPSP framework[23] predicts drug-drug interactions by constructing novel drug feature vectors using multiple types of drug information and Jaccard similarity, followed by a deep neural network for classification. Researchers[24] proposed NNPS, a neural network-based method for predicting polypharmacy side effects using novel drug feature vectors derived from mono side effects and drug-protein interactions.

Further we have analysed the correlated work of drug target interaction prediction problems. How the researchers have solved the drug target interaction problem. Neural Matrix Factorization (ANMF), a novel model for computational drug repositioning that integrates drug-drug and disease-disease similarities, autoencoder-based hidden feature extraction, and Generalized Matrix Factorization is porposed[25]. GraphCL-DTA, a model for drug-target binding affinity prediction that leverages graph contrastive learning to improve drug representation quality. GraphCL-DTA performs semantic-preserving data augmentation in the embedding space, allowing for richer, unsupervised representation learning of molecular graphs. Additionally, a new loss function is introduced to directly optimize the uniformity of drug and target embeddings[26]. Hybrid Attentional Memory Network (HAMN), a deep learning model that combines neighborhood-based and latent factor-based collaborative filtering approaches to leverage both local and global information for drug repositioning. By incorporating a memory-attention mechanism and an autoencoder, along with auxiliary drug and disease information, HAMN addresses the cold start problem[27]. Further researchers[28] present a multi-task

self-supervised learning framework for computational drug repositioning, designed to address the challenge of label sparsity by enhancing drug representation learning. By jointly training a main task for drug-disease association prediction and an auxiliary contrastive learning task using data augmentation. NMFDR[29], a Neural Metric Factorization model for computational drug repositioning, to overcome the expressive limitations of traditional matrix factorization methods. By modeling drug-disease associations through a generalized Euclidean distance in a high-dimensional latent space. The authors[30] propose PUON, it leverages only validated (positive) and unvalidated (unlabeled) drug-disease associations without negative sampling and introduces an Outer Neighborhood-based classifier to capture cross-feature information from latent factors. These various approaches demonstrate the growing importance of graph-based and neural network-driven methodologies in the prediction and understanding of drug interactions, reflecting an ongoing trend toward more advanced, interpretable, and accurate models in computational drug discovery.

Research gaps and challenges

Although significant progress has been made in drug-drug interaction prediction, several challenges and methodological limitations remain.

- **Data Imbalance and Incompleteness:** Drug interaction data usually come from medical experiments, patient records, literature reports, and departmental statistics. However, these data are often inconsistent. Furthermore, while the amount of drug-related data is vast, the subset available for analysis is relatively limited, leading to data imbalance. Many potential drug interactions remain undiscovered due to the scarcity of comprehensive datasets.
- **Complex Mechanisms of Interaction:** In the human body, drugs can interact during various phases such as metabolism, absorption, distribution, and excretion. These mechanisms often involve multiple biological processes, making it challenging to develop predictive models that can accurately capture and interpret such complex interactions.
- **Non-Euclidian relationship:** One of the primary obstacles is the vast complexity of interactions between drugs, which involves numerous factors such as chemical properties, pharmacodynamics, and pharmacokinetics. The large number of potential drug combinations further complicates the task, as traditional experimental methods for detecting DDIs through in vitro and in vivo testing are both time-consuming and costly. Additionally, many existing models rely heavily on structured data, such as chemical structures or biological pathways, but fail to fully capture the complex, unstructured relationships and nuances between drugs that can lead to interactions.
- **Data Fusion Challenges:** Effective drug-drug interaction analysis requires integrating multiple data sources, including multi-modal data like molecular structures, protein interaction networks, and genomic information. Combining these diverse data types to generate meaningful and useful features remains a significant challenge.
- **Data Sharing and Cross-Field Collaboration:** Predicting drug interactions often involves collaboration across diverse fields such as biology, chemistry, and medicine. However, issues related to data sharing, intellectual property rights, and the protection of sensitive information hinder cross-field cooperation, making it difficult to access and share crucial datasets.

Research significance

Theoretical Significance: Traditional drug-drug interaction prediction methods primarily rely on the chemical properties of drugs and target protein sequences, among other factors. However, predicting drug interactions remains a complex challenge. The recent rise of graph neural network (GNN) models offers a promising solution by enabling the modeling of intricate graph data, including the three-dimensional structures of drug molecules, chemical bond information, and target proteins. This approach provides richer and more precise feature representations for drug interaction prediction. Moreover, GNN models can capture relationships between drugs and target proteins, offering researchers deeper insights into the mechanisms of drug interactions. Accurate drug interaction predictions can also support personalized medicine, improving treatment outcomes and enhancing medication safety.

Practical Significance: The application of graph neural network (GNN) models has introduced considerable advantages for researchers using machine learning to predict drug interactions. GNN models significantly improve the accuracy of these predictions, positively influencing drug development, the therapeutic effectiveness of drug combinations, and drug safety. Therefore, using GNN models for drug interaction prediction presents vast research potential and practical implications for the field.

Results

Training results

The experiments are performed five times with different random seeds. Each model is trained for 100 epochs with learning rate=0.001, batch size $64 * 1024$, Hidden layer size=128,dropout rate=0.5.input-drop 0.0, edge-drop 0.0, attn-drop 0.0 and so on. We kept the default parameters given by the paper code itself for the baseline models.

The training loss curves (Figure 2) across the three datasets–BIOSNAP, DrugBank, and OGBL-DDI–exhibit consistent downward trends, indicating that all models generally learn meaningful patterns over epochs. In the BIOSNAP dataset, the models display smooth and monotonic convergence, with some achieving noticeably lower final losses. This suggests that BIOSNAP is relatively easier to fit, and some models generalize better on it. In contrast, the DrugBank dataset reveals greater variance in convergence behavior. Certain models converge rapidly, while others exhibit a slower decline or maintain higher loss values throughout training, possibly due to

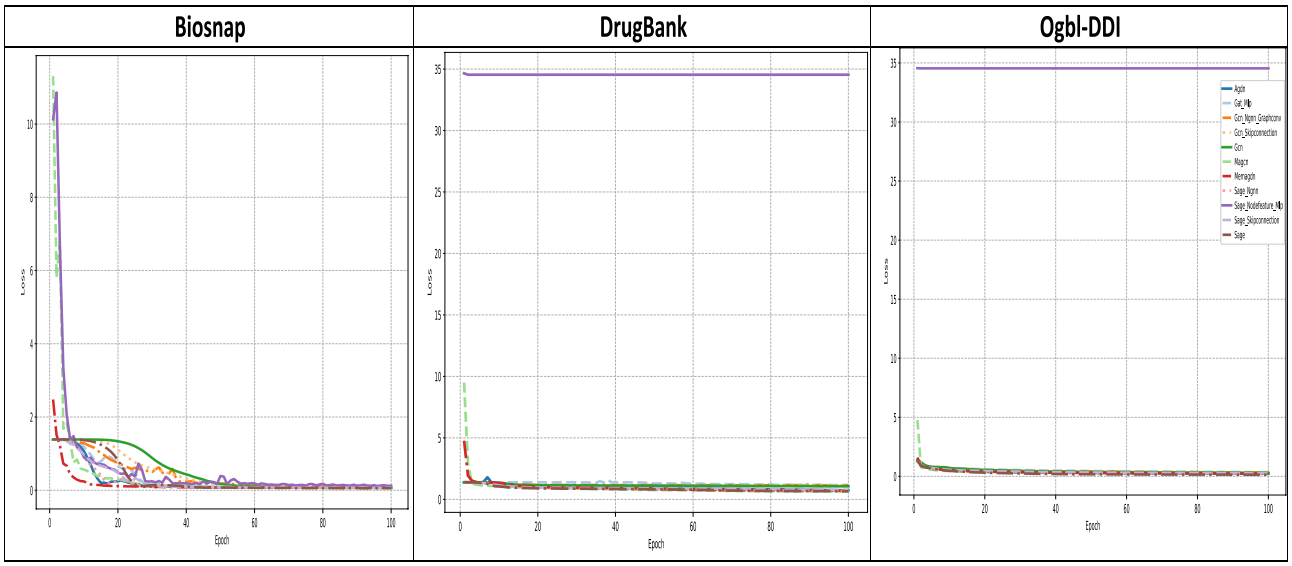


Fig. 2. The experimental results of loss value of each model during training for the three datasets. .

the dataset's increased complexity or noise. The OGBL-DDI dataset shows the most irregular loss trajectories, with several curves presenting mild fluctuations or delayed convergence. These patterns imply a more challenging optimization landscape, where model architecture significantly affects the training dynamics. Overall, the shape and rate of the loss curves reflect differences in model capacity, optimization stability, and dataset complexity.

Overview for the results on validation dataset Table 1 reports the highest validation performance of various models during training across the three datasets: *Biosnapddi*, *Drugbankddi*, and *Ogbl-ddi*, based on Hits@10 through Hits@50. The best three models per metric are color-coded, highlighting leading performance in ranking accuracy.

The validation results reveal distinct performance patterns across datasets. For `Biosnapddi`, models consistently achieve high `Hits@n` scores, indicating effective identification and ranking of true DDI pairs. Notably, 'AGDN', 'GCN', and 'MEMAGDN' demonstrate strong performance on this dataset. In contrast, all models exhibit very low `Hits@n` values on the `Drugbankddi` dataset, typically ranging from 0.010 to 0.030. This suggests a significant challenge in the models' ability to prioritize true DDI instances within their predictions for this specific dataset. 'MAGCN' shows a marginal advantage at higher 'n'. On the `Ogbl-ddi` dataset, model performance is moderate, yet more discerning. 'SAGE' and 'SAGENGNN' generally lead in 'Hits@n', with 'MAGCN' and 'AGDN' also performing competitively, achieving hit rates up to 0.630. In essence, while models effectively rank DDI pairs on `Biosnapddi`, they struggle considerably on `Drugbankddi`. The `Ogbl-ddi` dataset offers a more balanced evaluation ground, highlighting varying strengths among models in accurately retrieving relevant drug-drug interactions.

Overview from test data: Table 2 presents the final performance on test data of various models across three datasets: *Biosnapddi*, *Drugbankddi*, and *Ogbl-ddi*. The evaluation includes five ranking-based metrics, Hits@10 through Hits@50, along with overall classification accuracy. There is a clear distinction between the performance on Accuracy and Hits@n metrics. While many models achieve very high accuracy (often above 0.900), their Hits@n scores, particularly at lower 'n' values (e.g., @10, @20), are generally quite low, especially for the `Biosnapddi` and `Drugbankddi` datasets. This suggests that the models are highly accurate at classifying most DDI pairs (likely negative interactions, which are much more common), but struggle to rank true positive interactions highly within a set of predictions. In DDI prediction, a high Hits@n is often more crucial as it indicates the model's ability to prioritize relevant interactions.

The `Ogbl-ddi` dataset appears to be significantly more challenging for DDI prediction in terms of absolute Hits@n scores, but also shows a greater range and higher maximum Hits@n values, indicating that models have more room for differentiation and improvement on this dataset compared to `Biosnapddi` and `Drugbankddi`. The accuracy for `Ogbl-ddi` is also generally high, but slightly lower than `Biosnapddi`. For `Biosnapddi` and `Drugbankddi`, the Hits@n values are consistently close to 0.000 for most models and 'n' values. This indicates a substantial challenge in correctly ranking positive DDI instances among many negative ones.

Analysis by dataset

`Biosnapddi` dataset

- **Hits@n Performance:** This dataset presents a significant challenge for all models in terms of Hits@n. Most models show 0.000 ± 0.000 across all Hits@n metrics, up to @40 or even @50.

Dataset	Model	Hits@n				
		@10	@20	@30	@40	@50
Biosnapddi	AGDN	0.750 ± 0.020	0.890 ± 0.010	0.940 ± 0.010	0.970 ± 0.000	0.990 ± 0.000
Biosnapddi	GATMLP	0.620 ± 0.010	0.880 ± 0.010	0.940 ± 0.000	0.960 ± 0.000	0.980 ± 0.000
Biosnapddi	GCN	0.460 ± 0.010	0.740 ± 0.020	0.940 ± 0.010	0.980 ± 0.000	0.990 ± 0.000
Biosnapddi	GCNSKIPCONNECTION	0.360 ± 0.020	0.590 ± 0.010	0.900 ± 0.020	0.970 ± 0.000	0.980 ± 0.000
Biosnapddi	MEMAGDN	0.720 ± 0.020	0.900 ± 0.010	0.940 ± 0.010	0.970 ± 0.010	0.980 ± 0.000
Biosnapddi	MAGCN	0.680 ± 0.030	0.860 ± 0.010	0.940 ± 0.010	0.970 ± 0.010	0.980 ± 0.000
Biosnapddi	SAGE	0.780 ± 0.020	0.890 ± 0.010	0.930 ± 0.010	0.950 ± 0.010	0.960 ± 0.000
Biosnapddi	SAGENGNN	0.720 ± 0.010	0.890 ± 0.010	0.940 ± 0.010	0.970 ± 0.000	0.990 ± 0.000
Biosnapddi	SAGESKIPCONNECTION	0.740 ± 0.030	0.890 ± 0.010	0.930 ± 0.010	0.960 ± 0.000	0.980 ± 0.000
Drugbankddi	AGDN	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000	0.020 ± 0.000
Drugbankddi	GATMLP	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000
Drugbankddi	GCN	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000
Drugbankddi	GCNNGNNGRAPHCONV	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000
Drugbankddi	GCNSKIPCONNECTION	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000
Drugbankddi	MEMAGDN	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000
Drugbankddi	MAGCN	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000	0.030 ± 0.000
Drugbankddi	SAGE	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000
Drugbankddi	SAGENGNN	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000	0.020 ± 0.000
Drugbankddi	SAGESKIPCONNECTION	0.010 ± 0.000	0.010 ± 0.000	0.020 ± 0.000	0.020 ± 0.000	0.020 ± 0.000
Ogbl-ddi	AGDN	0.450 ± 0.040	0.520 ± 0.030	0.550 ± 0.030	0.570 ± 0.030	0.590 ± 0.020
Ogbl-ddi	GATMLP	0.250 ± 0.040	0.330 ± 0.050	0.370 ± 0.050	0.410 ± 0.050	0.440 ± 0.050
Ogbl-ddi	GCN	0.310 ± 0.020	0.380 ± 0.020	0.420 ± 0.010	0.440 ± 0.020	0.460 ± 0.010
Ogbl-ddi	GCNNGNNGRAPHCONV	0.300 ± 0.020	0.360 ± 0.020	0.400 ± 0.030	0.440 ± 0.030	0.470 ± 0.020
Ogbl-ddi	GCNSKIPCONNECTION	0.280 ± 0.030	0.340 ± 0.010	0.380 ± 0.010	0.420 ± 0.020	0.440 ± 0.020
Ogbl-ddi	MEMAGDN	0.430 ± 0.040	0.500 ± 0.020	0.530 ± 0.020	0.560 ± 0.010	0.580 ± 0.010
Ogbl-ddi	MAGCN	0.450 ± 0.010	0.520 ± 0.020	0.560 ± 0.020	0.590 ± 0.020	0.610 ± 0.010
Ogbl-ddi	SAGE	0.420 ± 0.050	0.500 ± 0.040	0.560 ± 0.030	0.600 ± 0.020	0.630 ± 0.010
Ogbl-ddi	SAGENGNN	0.470 ± 0.020	0.530 ± 0.010	0.560 ± 0.010	0.580 ± 0.010	0.600 ± 0.010
Ogbl-ddi	SAGESKIPCONNECTION	0.390 ± 0.010	0.450 ± 0.010	0.490 ± 0.010	0.520 ± 0.020	0.540 ± 0.020

Table 1. Validation Metrics during training.

  - **AGDN** is the only model to show a slight improvement, achieving 0.010 ± 0.000 for Hits@30, @40, and @50. This small positive value, while still low, indicates it might be marginally better at retrieving true positives than others.
  - **GATMLP** and **MEMAGDN** also show 0.010 ± 0.000 at Hits@50, with MEMAGDN also achieving it at Hits@30 and @40.
  - The extremely low Hits@n values across the board suggest that if DDI prediction is about ranking potential interactions, these models are not performing effectively on this dataset.

- **Accuracy Performance:** In contrast to Hits@n, accuracy is remarkably high for most models on `Biosnapddi`.

  - **AGDN** and **SAGESKIPCONNECTION** share the top accuracy at 0.992 ± 0.000, suggesting excellent performance in classifying DDI vs. non-DDI pairs.
  - **GCNSKIPCONNECTION** and **SAGENGNN** also perform very well with 0.991 ± 0.000 and 0.991 ± 0.001 respectively.
  - Models like **MEMAGDN** (0.891 ± 0.178) and **MAGCN** (0.888 ± 0.068) show significantly lower accuracy and higher standard deviations, indicating less stable or generally poorer classification performance compared to the others.

- On `Biosnapddi` dataset models are very good at classifying (leading to high accuracy). However, they struggle to correctly rank or identify the rare positive interactions (leading to very low Hits@n). For practical DDI prediction where identifying *true* interactions is paramount, the current models are not effectively addressing this challenge including our proposed models.

`Drugbankddi` dataset

- **Hits@n Performance:** Similar to `Biosnapddi`, the Hits@n scores for `Drugbankddi` are consistently very low (0.000 ± 0.000 or 0.010 ± 0.000 across all 'n' values for all models).

Dataset	Model	Hits@n					Accuracy
		@10	@20	@30	@40	@50	
Biosnapddi	AGDN	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.992 ± 0.000
Biosnapddi	GATMLP	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.989 ± 0.007
Biosnapddi	GCN	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.987 ± 0.001
Biosnapddi	GCNSKIPCONNECTION	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.991 ± 0.000
Biosnapddi	MEMAGDN	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.891 ± 0.178
Biosnapddi	MAGCN	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.888 ± 0.068
Biosnapddi	SAGE	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.980 ± 0.002
Biosnapddi	SAGENGNN	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.991 ± 0.001
Biosnapddi	SAGESKIPCONNECTION	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.992 ± 0.000
Drugbankddi	AGDN	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.798 ± 0.019
Drugbankddi	GATMLP	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.614 ± 0.070
Drugbankddi	GCN	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.736 ± 0.003
Drugbankddi	GCNNGNNGRAPHCONV	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.709 ± 0.021
Drugbankddi	GCNSKIPCONNECTION	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.500 ± 0.000
Drugbankddi	MEMAGDN	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.738 ± 0.097
Drugbankddi	MAGCN	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.699 ± 0.114
Drugbankddi	SAGE	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.801 ± 0.015
Drugbankddi	SAGENGNN	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.810 ± 0.030
Drugbankddi	SAGESKIPCONNECTION	0.000 ± 0.000	0.000 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.010 ± 0.000	0.807 ± 0.009
Ogbl-ddi	AGDN	0.050 ± 0.010	0.120 ± 0.010	0.150 ± 0.020	0.180 ± 0.020	0.230 ± 0.030	0.975 ± 0.003
Ogbl-ddi	GATMLP	0.090 ± 0.040	0.150 ± 0.040	0.200 ± 0.030	0.210 ± 0.020	0.250 ± 0.040	0.937 ± 0.011
Ogbl-ddi	GCN	0.090 ± 0.040	0.190 ± 0.020	0.260 ± 0.010	0.290 ± 0.050	0.350 ± 0.040	0.951 ± 0.003
Ogbl-ddi	GCNNGNNGRAPHCONV	0.140 ± 0.040	0.260 ± 0.040	0.330 ± 0.040	0.390 ± 0.040	0.440 ± 0.030	0.942 ± 0.007
Ogbl-ddi	GCNSKIPCONNECTION	0.160 ± 0.050	0.280 ± 0.030	0.340 ± 0.030	0.380 ± 0.020	0.410 ± 0.010	0.928 ± 0.007
Ogbl-ddi	MEMAGDN	0.060 ± 0.010	0.110 ± 0.020	0.150 ± 0.030	0.200 ± 0.020	0.240 ± 0.020	0.973 ± 0.002
Ogbl-ddi	MAGCN	0.120 ± 0.050	0.200 ± 0.040	0.230 ± 0.060	0.280 ± 0.060	0.350 ± 0.040	0.977 ± 0.001
Ogbl-ddi	SAGE	0.050 ± 0.010	0.090 ± 0.020	0.130 ± 0.020	0.150 ± 0.030	0.220 ± 0.010	0.981 ± 0.001
Ogbl-ddi	SAGENGNN	0.070 ± 0.030	0.090 ± 0.020	0.100 ± 0.030	0.140 ± 0.040	0.170 ± 0.040	0.972 ± 0.003
Ogbl-ddi	SAGESKIPCONNECTION	0.050 ± 0.020	0.120 ± 0.010	0.170 ± 0.020	0.210 ± 0.030	0.240 ± 0.030	0.964 ± 0.003

Table 2. Model evaluation based on various metrics and datasets.

  - Again, this indicates a struggle in ranking positive DDI interactions. All models show identical (or very nearly identical) Hits@n performance here.

- **Accuracy Performance:** Accuracy on `Drugbankddi` is generally lower than on `Biosnapddi` but still moderately high for several models.

  - **SAGENGNN** performs best in terms of accuracy at 0.810 ± 0.030.
  - **SAGESKIPCONNECTION** (0.807 ± 0.009) and **SAGE** (0.801 ± 0.015) also show strong accuracy.
  - **AGDN** (0.798 ± 0.019) is competitive.
  - **GATMLP** (0.614 ± 0.070) and especially **GCNSKIPCONNECTION** (0.500 ± 0.000) show notably poorer accuracy, with the latter essentially performing at random chance for binary classification.

- The `Drugbankddi` dataset also presents a challenge for ranking positive DDIs (evident from the consistently low Hits@n). While overall accuracy is respectable for some models, others perform poorly, suggesting a more balanced or diverse set of challenges compared to `Biosnapddi` in terms of classification. The significant drop in accuracy for certain models like GCNSKIPCONNECTION suggests it struggles with the underlying patterns in this specific dataset.

`Ogbl-ddi` dataset

- **Hits@n Performance:** This is where the models show much more meaningful differentiation and higher absolute Hits@n scores.

  - **GCNNGNNGRAPHCONV** is the standout performer in Hits@n, achieving the highest scores across most 'n' values: 0.140 ± 0.040 (@10), 0.260 ± 0.040 (@20), 0.330 ± 0.040 (@30), and the highest at 0.390 ± 0.040 (@40) and 0.440 ± 0.030 (@50). This indicates it is the most effective at retrieving true positive DDIs within the top ranked predictions.
  - **GCNSKIPCONNECTION** is also very strong, particularly at Hits@10 (0.160 ± 0.050 - highest) and @20 (0.280 ± 0.030 - highest). Its performance slightly drops at higher 'n' compared to GCNNGNNGRAPHCONV.
  - Other models like **GCN** (e.g., 0.350 ± 0.040 at @50) and **MAGCN** (e.g., 0.350 ± 0.040 at @50) also show decent Hits@n.
  - **SAGE**, **SAGENGNN**, and **AGDN** generally have lower Hits@n values on this dataset compared to the top performers.

- **Accuracy Performance:** Accuracy is high across most models on `Ogbl-ddi`.
  - **SAGE** achieves the highest accuracy at 0.981 ± 0.001.
  - **MAGCN** (0.977 ± 0.001) and **MEMAGDN** (0.973 ± 0.002) also show excellent accuracy.
  - **AGDN** (0.975 ± 0.003) is also among the top.
  - **GCNSKIPCONNECTION** (0.928 ± 0.007) and **GATMLP** (0.937 ± 0.011) have slightly lower accuracy but are still performing well.
- The `Ogbl-ddi` **:** dataset appears to be better suited for evaluating the ranking capabilities of DDI models. Models like **GCNNGNNGRAPHCONV** and **GCNSKIPCONNECTION** demonstrate superior ability in retrieving actual DDI instances, indicating their potential for real-world DDI prediction where identifying relevant interactions is key. High accuracy is maintained by most models, suggesting a good overall classification ability.

Analysis by model (across datasets)

AGDN

- **Hits@n:** Consistently low on `Biosnapddi` and `Drugbankddi` (0.000-0.010). Shows moderate performance on `Ogbl-ddi` (e.g., 0.230 at @50).
- **Accuracy:** Excellent on `Biosnapddi` (0.992), decent on `Drugbankddi` (0.798), and strong on `Ogbl-ddi` (0.975).
- **Overall:** A generally robust performer in terms of accuracy, but its Hits@n performance is only notable on `Ogbl-ddi` among the datasets tested.

GATMLP

- **Hits@n:** Very poor on `Biosnapddi` and `Drugbankddi` (mostly 0.000-0.010). Moderate on `Ogbl-ddi` (e.g., 0.250 at @50).
- **Accuracy:** Good on `Biosnapddi` (0.989), but significantly lower on `Drugbankddi` (0.614) and slightly lower on `Ogbl-ddi` (0.937) compared to top models.
- **Overall:** Seems to struggle with `Drugbankddi` dataset in particular. Hits@n are not its strong suit.

GCN

- **Hits@n:** Very low on `Biosnapddi` and `Drugbankddi`. Better on `Ogbl-ddi` (e.g., 0.350 at @50), performing reasonably well among the models.
- **Accuracy:** High on `Biosnapddi` (0.987), moderate on `Drugbankddi` (0.736), and good on `Ogbl-ddi` (0.951).
- **Overall:** A solid baseline GNN model, showing average to good performance, particularly for Hits@n on `Ogbl-ddi`.

GCNSKIPCONNECTION

- **Hits@n:** Very low on `Biosnapddi` and `Drugbankddi`. **Very strong on** `Ogbl-ddi`, particularly at lower 'n' values (highest Hits@10 and @20).
- **Accuracy:** Excellent on `Biosnapddi` (0.991), but remarkably poor on `Drugbankddi` (0.500), and decent on `Ogbl-ddi` (0.928).
- **Overall:** A peculiar model. While it excels at retrieving relevant hits on `Ogbl-ddi`, its highly variable accuracy across datasets (especially the random performance on `Drugbankddi`) suggests it might be sensitive to specific dataset characteristics or require more fine-tuning.

MEMAGDN

- **Hits@n:** Marginally better than GCN on `Biosnapddi` (0.010 at @30,@40,@50) and consistent with others on `Drugbankddi` (0.010). Moderate on `Ogbl-ddi` (e.g., 0.240 at @50).
- **Accuracy:** Noticeably lower accuracy on `Biosnapddi` (0.891) and `Drugbankddi` (0.738) with higher variance, but high on `Ogbl-ddi` (0.973).
- **Overall:** Its performance is quite inconsistent across datasets, suggesting it might not generalize as well as some other models.

MAGCN

- **Hits@n:** Very low on `Biosnapddi` and `Drugbankddi`. Decent on `Ogbl-ddi` (e.g., 0.350 at @50).
- **Accuracy:** Relatively low and variable accuracy on `Biosnapddi` (0.888) and `Drugbankddi` (0.699), but excellent on `Ogbl-ddi` (0.977).
- **Overall:** Similar to MEMAGDN, its performance varies significantly, showing strong accuracy on `Ogbl-ddi` but struggling on the other two in terms of accuracy.

SAGE

- **Hits@n:** Very low on `Biosnapddi` and `Drugbankddi`. Lower end on `Ogbl-ddi` (0.220 at @50) compared to top models.
- **Accuracy:** High on `Biosnapddi` (0.980), highest on `Drugbankddi` among models with 0.801, and highest on `Ogbl-ddi` (0.981).
- **Overall:** SAGE is consistently among the top models for accuracy across all datasets, indicating its strong classification ability. However, its Hits@n performance is generally not as competitive as GCNNGNNGRAPHCONV or GCNSKIPCONNECTION on `Ogbl-ddi`.

SAGENGNN

- **Hits@n:** Very low on `Biosnapddi` and `Drugbankddi`. Lower end on `Ogbl-ddi` (0.170 at @50).
- **Accuracy:** High on `Biosnapddi` (0.991), highest on `Drugbankddi` (0.810), and high on `Ogbl-ddi` (0.972).
- **Overall:** Similar to SAGE, it excels in overall classification accuracy across all datasets, but its ability to rank true positives highly (Hits@n) is not its strongest point.

SAGESKIPCONNECTION

- **Hits@n:** Very low on `Biosnapddi` and `Drugbankddi`. Lower end on `Ogbl-ddi` (0.240 at @50).
- **Accuracy:** Top accuracy on `Biosnapddi` (0.992) and strong on `Drugbankddi` (0.807) and `Ogbl-ddi` (0.964).
- **Overall:** A very strong performer in terms of accuracy across all datasets, similar to SAGE and SAGENGNN. Its Hits@n performance on `Ogbl-ddi` is slightly better than SAGE/SAGENGNN but still not as good as GCNNGNNGRAPHCONV or GCNSKIPCONNECTION.

Discussion

Accoding to experimental results based on various models, the analysis reveals that while many models achieve high accuracy in classifying drug-drug interaction (DDI) pairs, they struggle significantly with ranking true positive interactions, particularly on the `biosnapddi` and `drugbankddi` datasets. The `ogbl-ddi` dataset, however, shows more differentiation and higher Hits@n scores, indicating a better environment for evaluating ranking capabilities. Models like `GCNNGNNGRAPHCONV` and `GCNSKIPCONNECTION` stand out for their performance in Hits@n on `ogbl-ddi`, while `SAGE`, `SAGENGNN`, and `SAGESKIPCONNECTION` excel in accuracy across all datasets. We have found that not single models outperforms on all the metrics on the datasets which shows the DDI dataset have inherent different characteristics which one model doesn't fit on all the datasets.This experimental result will provide important references and guidance for drug interaction prediction using graph machine learning, significantly contributing to the sustainable development of drug interaction research and offering valuable insights for real-life drug use. In future research, we can consider the following aspects:

1. Use different models to conduct experiments on various datasets and leverage the characteristics of different models to combine them for discovering better drug interaction prediction models.
2. Actively collaborate with experts in biology and medicine to update and optimize models based on relevant clinical trial data and pharmacological knowledge.
3. Investigate methods to improve the interpretability and explainability of the model so that researchers in related fields can understand the model's prediction results.

Methods

Graph neural network models

GAT model

The Graph Attention Network (GAT) represents an advanced graph neural network model designed to leverage the attention mechanism for capturing intricate relationships among nodes within a graph, ultimately producing comprehensive node representations[31]. Unlike traditional methods, GAT introduces an innovative approach by assigning adaptive attention weights to neighboring nodes, allowing the model to focus on the most relevant connections. This mechanism enables efficient learning without relying on computationally intensive operations, such as matrix inversion, or requiring prior knowledge of the graph's structure. By incorporating multiple stacked layers, GAT systematically aggregates domain-specific features from the local neighborhood of each node. The detailed architecture of the GAT model is depicted in Figure 3(a).

GCN model

The Graph Convolutional Network (GCN) is a deep learning model designed for processing graph-structured data. It is primarily utilized for tasks such as node classification and link prediction[32]. GCN represents graph data using two key components: an adjacency matrix and a node feature matrix. The adjacency matrix encodes the relationships between nodes, where entries typically indicate the presence (1) or absence (0) of connections, while the node feature matrix contains the feature vectors for each node. Through graph convolution operations, GCN propagates and aggregates node features from neighboring nodes, thereby generating enriched node representations. These representations are iteratively refined across multiple layers of graph convolutions, enabling the model to capture both local connectivity and complex relationships. GCN supports parameter sharing and is

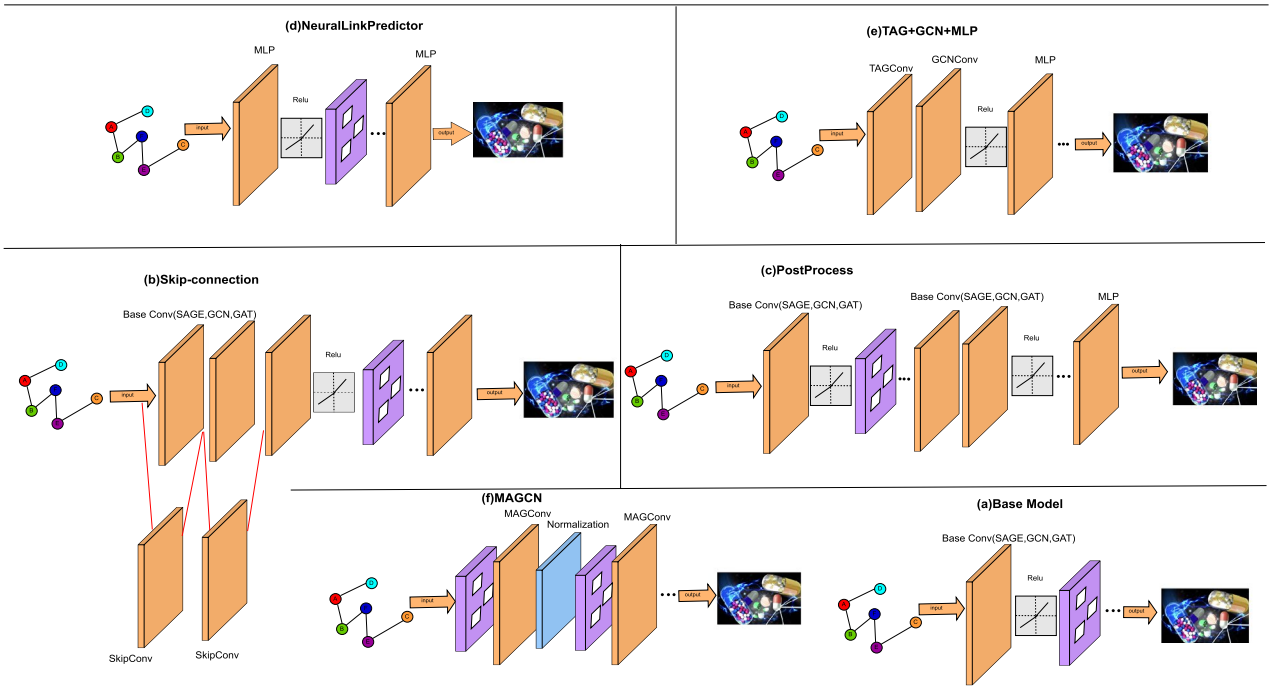


Fig. 3. Various graph neural networks used in our study are described in this figure.

adaptable to graphs of various structures, providing strong generalization capabilities. Furthermore, GCNs can be integrated with other deep learning models to enhance performance across diverse tasks. The architecture of the GCN model is illustrated in Figure 3(a).

SAGE model
The SAGE (Sample and Aggregation) model is a graph neural network model developed to learn low-dimensional representations of nodes in graph data. It effectively captures both local and global characteristics of nodes through multilayer neighbor sampling and information aggregation operations[33]. It is depicted in Figure 3(a).

Graph Sage + NGNN (SAGENGNN)
In this model, the GraphSAGE layer and the Nested Graph Neural Network (NGNN) layers are combined sequentially to enhance feature extraction, following a linear architecture inspired by[34].

TAG model
The Topology Adaptive Graph (TAG) method adapts the topology of graph neural networks to suit the structure of different graph data[35]. Traditional graph neural networks often assume a fixed topology, which may not be ideal for diverse graph structures. TAG dynamically adjusts the network's topology to fit the input graph data better and capture local structural information better. It allows each node to adaptively select its neighboring nodes and learn convolutional kernels based on the relationships between nodes. This flexibility enables TAG to better capture local structural information, enhancing the performance of graph neural networks across different types of graph data.

To tackle the drug-drug interaction problem, we explored various combinations of these models, systematically evaluating their performance and comparing the results with those of other state-of-the-art approaches.

Memory attention graph convolution(MAGCN):

Wang et al.[36] introduced the Graph Inception Diffusion Networks (GIDN) model, which extends graph diffusion mechanisms across diverse feature spaces while incorporating an inception module to mitigate the computational overhead associated with complex network structures. The architecture of the GIDN model is illustrated in Figure 3(f). The primary components of the model are as follows:

Feature transformation
Node features ($\mathrm{X} \in \mathbb{R}^{N \times F_{\mathrm{in}}}$) are first split into smaller blocks corresponding to hop steps. For each block, a transformation is applied. The feature transformation for a block *i* is defined as:

$$\mathrm{X}^{(i)}_{\mathrm{transformed}} = \mathrm{X}^{(i)} + \mathrm{P}^{(i)}$$

where $\mathrm{P}^{(i)} \in \mathbb{R}^{1 \times H \times F_{\mathrm{in}}/S}$ is a learned position embedding for the *i*-th hop.

Attention mechanism
The layer computes attention between nodes in a graph. The attention mechanism uses source (q) and destination (k) node features for computing attention scores.

Query, Key, and Value matrices are used to transform node features:

$$\mathrm{Q} = \mathrm{W}_Q\mathrm{X}, \quad \mathrm{K} = \mathrm{W}_K\mathrm{X}, \quad \mathrm{V} = \mathrm{W}_V\mathrm{X}$$

where $\mathrm{W}_Q, \mathrm{W}_K, \mathrm{W}_V \in \mathbb{R}^{F_{\text{in}} \times F_{\text{out}}}$ are the learned projection matrices.

The attention score computation for each edge is given by:

$$\alpha_{ij} = \text{LeakyReLU}\left(\mathrm{a}^\top \left[\mathrm{Q}_i \| \mathrm{K}_j\right]\right)$$

where $\mathrm{a} \in \mathbb{R}^{2F_{\text{out}}}$ is a learned vector, and $\|$ denotes concatenation. The attention scores are normalized using softmax:

$$\alpha_{ij} = \frac{\exp(\alpha_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(\alpha_{ik})}$$

where $\mathcal{N}(i)$ denotes the set of neighboring nodes of node *i*.

The final node features are aggregated using attention weights:

$$\mathrm{h}_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathrm{V}_j$$

Edge features (Optional)
If edge features ($\mathrm{E} \in \mathbb{R}^{E \times D_e}$) are provided, they are incorporated into the attention mechanism. Specifically, the edge features modify the attention weights as:

$$\alpha_{ij} = \frac{\exp(\alpha_{ij} + \mathrm{e}_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(\alpha_{ik} + \mathrm{e}_{ik})}$$

where $\mathrm{e}_{ij} \in \mathbb{R}^{D_e}$ represents the edge feature vector between nodes *i* and *j*.

Multi-hop neighborhood aggregation
The model aggregates features over multiple hop steps. It updates node features and computes attention for each hop, allowing information to flow over longer distances.

TAG and Jumping Knowledge Layer
$\mathrm{H} = \mathrm{JK}(\mathrm{TAGConv}(\mathrm{H}),\ \mathrm{aggr} = \mathrm{Maxpooling})$

Output and residual connection
After attention aggregation, the features are transformed through a final linear transformation. The output is:

$$\mathrm{H}_{\text{out}} = \mathrm{W}_{\text{out}} \cdot \text{Concat}\left(\mathrm{H}^{(1)}, \mathrm{H}^{(2)}, \ldots, \mathrm{H}^{(S)}\right) \tag{1}$$

If a residual connection is used, the input features are added back to the output, potentially via a transformation matrix:

$$\mathrm{H}_{\text{out}} = \mathrm{H}_{\text{out}} + \mathrm{W}_{\text{res}}\mathrm{X} \tag{2}$$

Adaptive Graph Diffusion Networks (AGDN)[37]
Sun et al.[37] proposed Adaptive Graph Diffusion Networks (AGDNs) to address overfitting, over-smoothing, and inefficiency in deep Graph Neural Networks. AGDNs perform generalized multi-hop diffusion across different feature spaces using learnable hop-wise weighting mechanisms–Hop-wise Attention and Hop-wise Convolution–while maintaining moderate runtime and complexity. They have also given a variant MEMAGDN.

Model optimization

Model optimization methods[38]

1. **Skip Connection:** Skip connection is a technique employed in neural networks to establish direct connections between different layers. It functions by adding the output of one layer directly to the input of another, enabling information to bypass intermediate layers and propagate more effectively through the network. This approach alleviates the vanishing gradient problem and facilitates the training of deeper networks. Moreover, skip connections allow lower-level features to be retained and utilized by higher layers. In our experiments, as illustrated in Figure 3(b), we incorporated skip connections into the basic model. This enhancement enables the model to aggregate information from neighbors over multiple hops while mitigating the risk of over-smoothing in node embeddings.

2. **Post-Processing Layers:** While adding more convolutional layers can be challenging in GNNs, it is also possible to incorporate pre- and post-processing layers that do not perform message passing. Post-processing layers, such as multi-layer perceptrons (MLPs), are applied to node embeddings to convert and refine the nonlinear outputs of the model. MLPs consist of fully connected layers containing multiple neurons and using nonlinear activation functions such as ReLU or Sigmoid. Figure 3(c) illustrates the model structure with these post-processing layers.
3. **Enhanced Predictor:** Building on the idea that passing embeddings through MLP layers can enhance performance, we incorporate this concept into a GNN model. Instead of integrating the LinkPredictor as part of the GNN, we define it as a small neural network ("NeuralLinkPredictor") to perform post-processing independently. This setup optimizes the model's predictive abilities, as depicted in Figure 3(d).

Model construction

The AGDN component is a specialized neural network designed for processing graph data by aggregating information from neighboring nodes through diffusion operations. By iteratively updating node representations based on their neighbors, the AGDN effectively captures the graph's structural information, making it well-suited for understanding complex relationships in graph data.

Despite its strengths, the AGDN alone may face limitations in addressing the dynamic and intricate nature of graph topologies encountered in real-world applications. To address this, the TAG model introduces a topology-adaptive mechanism that dynamically adjusts to the graph's structural variations based on the data's characteristics. This enables the TAG model to better capture essential topological features, enhancing prediction accuracy.

In the MAGCN model, the AGDN, Attention based Neibhour aggregation, and TAGConv layer is applied to the node features, further refining the learned representations by adaptively accounting for the graph's topology. The final component, the MLP, performs classification or regression tasks on the updated node features. As a robust nonlinear mapping tool, the Jumping Knowledge layer transforms these features into the target output space, enabling accurate and reliable predictions.

In summary, the MAGCN model effectively combines the complementary strengths of AGDN, Attention mechanism, TAG, and Jumping Knowledge layer to achieve efficient graph data processing and precise predictive performance. This integrated approach offers wide applicability and has demonstrated strong results in practical applications.

Model training and optimization process

Problem formulation

During the initial training phase, negative sampling is employed to address the imbalance between positive and negative edges in the graph. For each positive edge (i.e., an existing edge in the training graph), an equal number of negative edges (i.e., non-existent edges) is generated. This approach ensures that each training batch contains a balanced representation of positive and negative edges, facilitating effective learning.

The model then generates node embeddings by processing all training edges, transforming the graph data into feature representations that encapsulate the structural and relational information of the graph. These embeddings serve as the input for subsequent prediction tasks.

As a result, the problem is formulated as a binary classification task, where the objective is to distinguish between positive and negative edges based on their learned feature representations. This framing aligns with standard machine learning methodologies, allowing the model to leverage well-established classification techniques for edge prediction.

Choice of loss function

The experiment employs binary cross-entropy loss to evaluate model performance. This loss function is particularly suited for binary classification problems, where it measures the discrepancy between two probability distributions: one representing the true labels and the other representing the model's predictions.

In the context of binary classification, each sample is assigned a binary label, typically denoted as 0 or 1, where 0 represents the negative class and 1 denotes the positive class. The model generates a probability value between 0 and 1, indicating the likelihood that a given sample belongs to the positive class. The binary cross-entropy loss evaluates the model's performance by comparing the predicted probability with the actual true label. The loss function can be mathematically expressed as follows:

$$Loss = -\frac{1}{N}\sum_{i=1}^{N} h_i \cdot \log(f(h_i)) + (1 - h_i) \cdot \log(1 - f(h_i)) \tag{3}$$

Among them, h is the binary label 0 or 1, and $f(h)$ is the predicted probability that the output belongs to the h label.

Optimizer selection and parameter settings

The experiment employs the Adam optimizer for model training. Adam, which stands for Adaptive Moment Estimation, is an adaptive learning rate optimization algorithm that combines the benefits of momentum-based gradient descent with the advantages of adaptive learning rates[39]. It adjusts the learning rate of each parameter by calculating both the first-order moment estimate (the mean) and the second-order moment estimate (the uncentered variance) of the gradients. This adaptive mechanism allows Adam to effectively handle sparse gradients and noisy data.

Evaluation indicators

The experiment employs two performance evaluation metrics: Hits@K and Accuracy. The Hits@K metric is designed to rank true drug interactions higher than non-interacting drug pairs, emphasizing the model's ability to correctly identify relevant interactions. Hits@K is particularly useful in ranking tasks, where the goal is to ensure that the true positive interactions are within the top K predictions made by the model. The specific calculation process is as follows:

1. Hits@K:

The process begins by calculating the rank of each positive sample based on the predicted scores. If a positive sample's rank is less than or equal to K, it is considered a "hit" and contributes to the HIT value. Finally, the HIT value is computed by calculating the ratio of the number of hit positive samples to the total number of positive samples, M, as follows:

$$\text{HIT@K} = \frac{\text{Number of hit positive samples}}{M} \tag{4}$$

$$Hits@K\{positive\} = \frac{\{\text{NumberofpositivesampleswithRank} \leq K\}}{M} \tag{5}$$

Next, for each negative sample, its ranking in the predicted scores is calculated. If the rank is greater than K, the negative sample is considered a "hit" and contributes to the HIT value. The HIT value for the negative samples is then computed by calculating the ratio of the number of hit negative representative data to the total number of negative representative data ($N - M$), as follows:

$$Hits@K\{negative\} = \frac{\{\text{NumberofnegativesampleswithRank} > K\}}{N - M} \tag{6}$$

The final HIT value (Hits@K) is the average of the positive HIT value and the negative HIT value:

$$Hits@K = \frac{Hits@K\{positive\} + Hits@K\{negative\}}{2} \tag{7}$$

Among them, it is assumed that there are N samples, of which M are positive samples and $N - M$ are negative samples. "Number of positive samples with Rank $\leq$ K" refers to the number of positive samples with predicted score rankings less than or equal to K. "Number of negative samples with Rank $\geq$ K" refers to the number of negative samples with predicted score rankings greater than or equal to K.

2. Accuracy: Accuracy is another widely used evaluation metric in classification tasks, which measures the proportion of samples that are correctly classified by the model. The formula for calculating accuracy is as follows:

$$Accuracy = \frac{\{Numberofcorrectlypredictedsamples\}}{N} \tag{8}$$

where "Number of Correct Predictions" represents the number of samples correctly classified by the model, and N represents the total number of samples.

Data set selection

In this study, we have used three publicly available drug-drug interaction datasets. These datasets have their own strengths and weaknesses, making them altogether great choices for evaluating models' performance. The datasets are as follows:

1. **OGB-DDI:** The Ogbl-ddi dataset is a homogeneous, unweighted, undirected graph where each node corresponds to an FDA-approved or experimental drug. Edges denote drug-drug interactions, capturing cases where the combined effect of two drugs differs significantly from their independent effects[40,41]. The dataset comprises 4,267 nodes and 1,067,911 edges, with an average node degree of 501. This structure enables the exploration of complex drug-drug interactions.
2. **Drugbank-DDI:**The DrugBank drug-drug interaction (DDI) dataset is manually curated from FDA and Health Canada drug labels, along with primary literature sources. It includes 191,808 drug-drug interaction pairs involving 1,706 unique drugs and spans 86 distinct interaction types. The associated task is formulated as a multi-class classification problem: given the SMILES representations of two drugs, the goal is to predict the type of interaction between them[40,42].
3. **Biosnap-DDI:**The BIOSNAP dataset comprises 1,322 FDA-approved drugs with 41,520 labeled drug-drug interactions (DDIs). In contrast, the Zhang-DDI dataset includes 548 drugs and 48,584 pairwise DDIs. Both datasets are used for multi-DDI prediction tasks, where the objective is to identify specific interaction types between drug pairs[43].

Dataset characteristics

In Figure 4 we have shown ddi interaction adjacency plot for Ogbl-ddi dataset. In Figure 5 we have shown the label distribution for DrugBank ddi and Biosnap DDI. The figure shows these these are balanced class. For Ogbl-ddi dataset only interaction graph were present. To make it balanced we have done global sampling with equal number of true class for training the data. In this section we have anlysed various characteristics of our datasets.

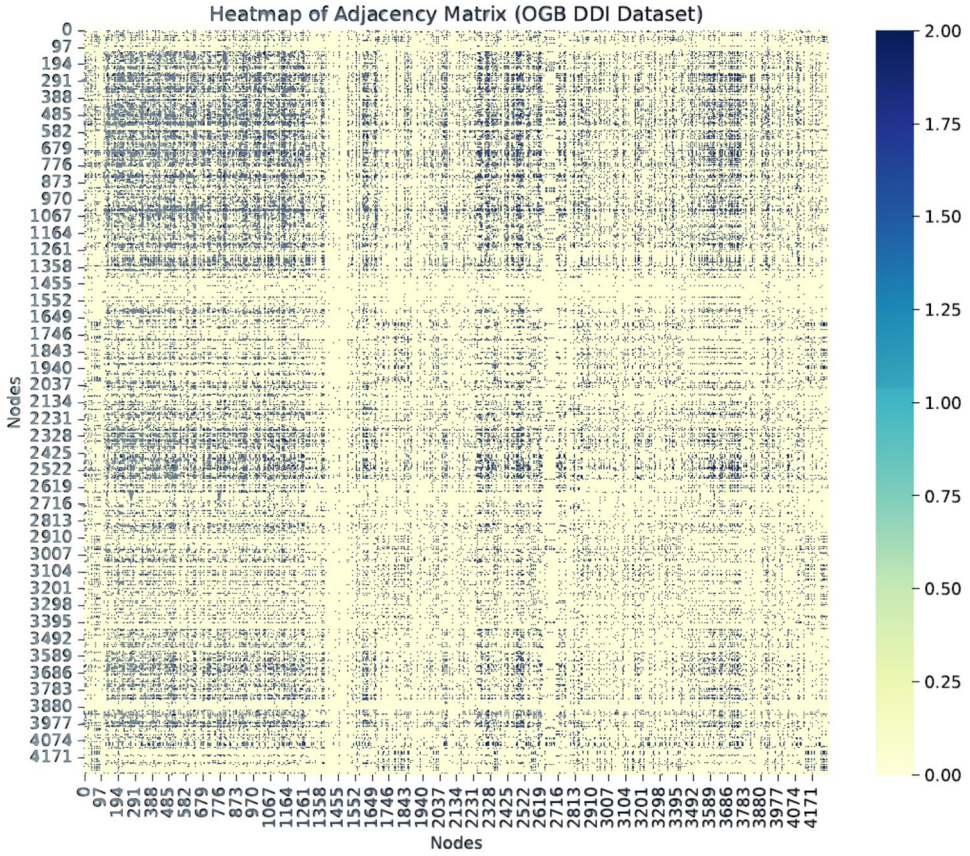


Fig. 4. Whole (un-directed) graph adjacency matrix plot. Each node represents a drug. The dots show there is an interaction between two drugs.

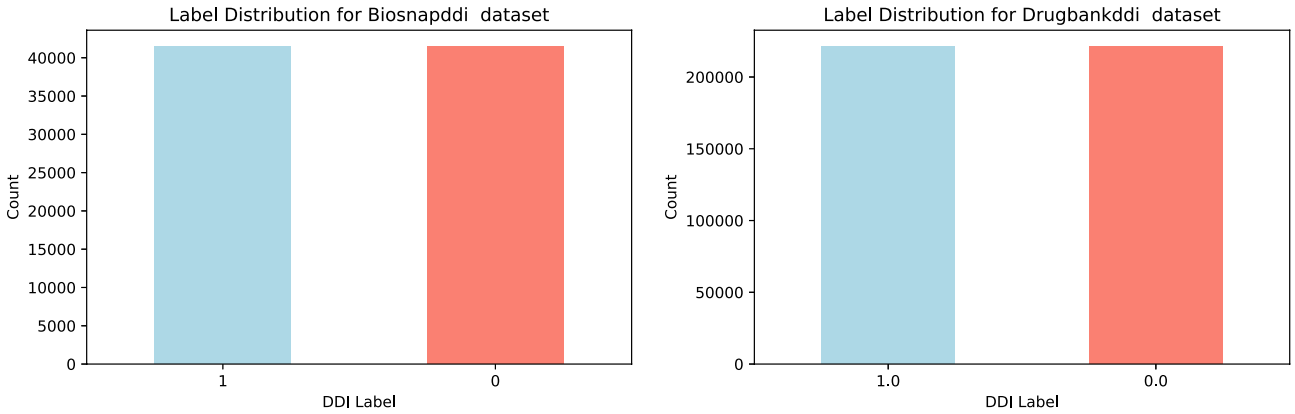


Fig. 5. This is label distribution of both the datasets. Which means both datasets are balanced.

Although in the following analysis we have only considered two datasets because the have provided drug smiles in their datasets.

Tonimoto similarity

Tanimoto similarity, is a widely used metric in cheminformatics and drug discovery to quantify the structural similarity between two molecules. Once two molecules are represented by their fingerprints (let's call them A and B), the Tanimoto coefficient (T_c) is calculated using the following formula:

$$T_c(A, B) = \frac{N_{AB}}{N_A + N_B - N_{AB}}$$

Where:

- N_{AB} is the number of bits (features) that are "on" (present) in *both* fingerprint A and fingerprint B.
- N_A is the total number of "on" bits in fingerprint A.
- N_B is the total number of "on" bits in fingerprint B.

The Tanimoto coefficient ranges from 0 to 1, where:

- 1 indicates perfect structural similarity (the fingerprints are identical).
- 0 indicates no shared features (the fingerprints have no "on" bits in common).

Impact on Drug-Drug Interaction (DDI)
The core principle behind using Tanimoto similarity in these contexts is the **"similarity principle"**: *structurally similar molecules are likely to have similar properties and biological activities.*

- **Prediction of DDIs:** Tanimoto similarity is a key tool in predicting potential drug-drug interactions. The assumption is:
  - If Drug A interacts with Drug B, and Drug C is structurally very similar to Drug A, then Drug C is also likely to interact with Drug B.
  - Similarly, if two drugs have similar molecular structures, they might compete for the same metabolic enzymes (e.g., CYP enzymes) or transporters, leading to altered concentrations and effects of one or both drugs.
- **Mechanisms:** DDIs can arise from various mechanisms (pharmacokinetic or pharmacodynamic). Structural similarity can shed light on:
  - **Pharmacokinetic DDIs:** If two drugs are structurally similar, they might be metabolized by the same enzymes or transported by the same proteins, leading to competitive inhibition, induction, or altered clearance, which can increase or decrease drug levels in the body.
  - **Pharmacodynamic DDIs:** While less direct than pharmacokinetic interactions, highly similar drugs might have similar off-target effects or bind to the same receptors, leading to additive or antagonistic effects.

In Figure 6 we have shown the Tonimoto similarity index for both the datasets.

Molecular weight distribution

The Figure 7 shows molecular weight distribution for both the datasets. Molecular weight affects drug-drug and drug-target interactions by influencing absorption, distribution, binding affinity, and permeability. Generally, higher molecular weight can reduce membrane permeability, while also affecting how tightly or selectively a drug binds to its target or interacts with other drugs.

Average hydrophobicity (logP) of each drug pair

Molecular hydrophobicity affects drug-drug and drug-target interactions by influencing binding affinity, membrane permeability, and solubility. Hydrophobic drugs often bind more strongly to hydrophobic pockets on targets but may have poor solubility and higher chances of off-target or nonspecific interactions. In Figure 8 we have shown the average hydrophobicity for two datasets.

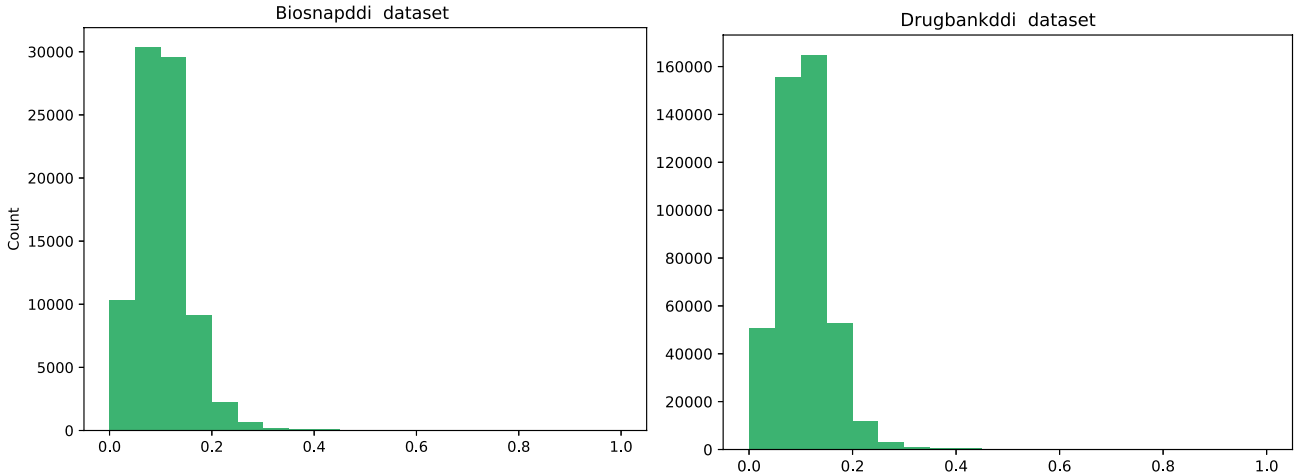


Fig. 6. Tonimoto similarity between drugs for both the datasets.

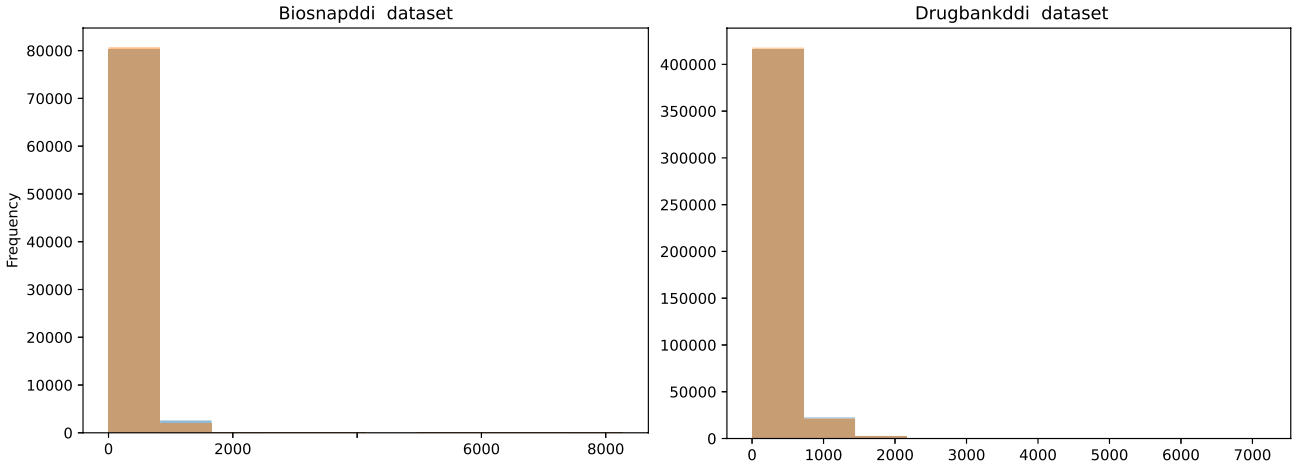


Fig. 7. Molecular weight distribution for both the datasets.

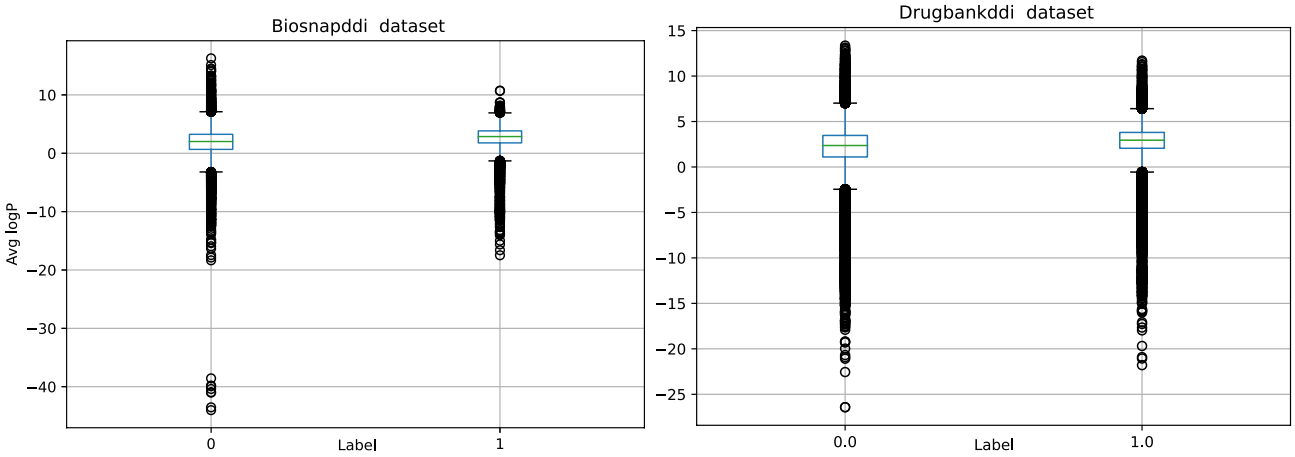


Fig. 8. Average hydrophobicity of each drug pair.

Data set partitioning

The drug edges in the dataset are segmented based on the specific proteins they target in the body. As a result, the test set consists of drugs that primarily bind to proteins distinct from those in the training and validation sets. This segmentation ensures that the drugs in the test set exhibit different biological mechanisms of action and interact with distinct proteins compared to those in the training and validation sets. Protein-target segmentation, therefore, provides an opportunity to assess the model's ability to make meaningful and generalizable predictions that are not limited to the specific proteins seen during training[41].

Experimental tasks

The experimental task focuses on predicting drug-drug interactions (DDIs) by ranking each known positive interaction against approximately 100,000 randomly sampled negative interactions. The primary evaluation metrics are Hits@10, Hits@20, and Hits@50, which measure the proportion of true positive interactions ranked within the top K predictions. These metrics assess the model's ability to correctly prioritize true interactions among a large set of candidates. In addition to Hits@K, we also report accuracy to provide a broader view of model performance across the entire prediction space. Prior studies have indicated that evaluating performance at $K = 20$ is particularly meaningful in DDI prediction tasks, but we extend this analysis to smaller and larger K values for a more comprehensive evaluation.

Experimental environment

The experimental code environment is based on Python version 3.10.8 and PyTorch version 2.1.2+cu118. The version of `torch-scatter` is 2.1.2+pt21cu118, `torch-cluster` is 1.6.3+pt21cu118, the version of `torch-sparse` is 0.6.18+pt21cu118, `torch-geometric` is 2.6.0, ogbl is 1.3.6, dgl is `2.4.0+cu118`, and `dgl-life` is 0.3.2.

Data availability

All the data are publically available and described as below: The online dataset can be downloaded using python obg package. The further information is provided on Stanford University website: https://ogb.stanford.edu/docs/linkprop/#ogbl-ddi The datasets used and/or analyzed during the current study are available in the OGB Repository. The tutorial how to use the data can be found here: https://ogb.stanford.edu/docs/home/

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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